

Madhav Singh Bhanot

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SUMMARY

Software Engineer with technical experience at **Innate Ai** and **KrishLabs**, specializing in full-stack engineering and AI/LLM integration, including OpenAI and Anthropic API integration, prompt engineering, and **RAG** pipelines with **React.js**, **Node.js**, **Express.js**, and databases including **PostgreSQL** and **MongoDB**. Proficient in responsive UIs and integrating **REST APIs** and **AI services** to deliver scalable product solutions. Awarded **3rd place** in a university-wide hackathon (200+ participants) for a **React Native** safety application; focused on impactful, end-to-end software systems.

EDUCATION

Bharati Vidyapeeth Deemed University | Pune, India |

Graduated 2026

Bachelor of Technology in Computer Science and Engineering | **CGPA: 8.32/10**

SKILLS

Languages: JavaScript, TypeScript, Java, SQL, HTML5, CSS3

Frontend: React, Next.js, React Native (Expo), Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, WebSockets, Socket.IO, Prisma ORM, JWT, Mongoose ORM

Databases: PostgreSQL, MongoDB, Firebase Firestore, Redis, Supabase

Cloud & Tools: AWS (EC2, S3), Git, Docker, CI/CD, Jira, Postman, Jest

Data & Workflow: Apache Airflow, Celery, ETL Pipelines, Web Scraping, MS Word

AI & LLM: OpenAI & Anthropic API Integration, Prompt Engineering, RAG, LLM Workflow Systems

EXPERIENCE

Innate AI | React Developer (Internship) | Helsinki, Finland Remote |

Jul 2025 – Oct 2025

- Architected production **React + TypeScript** dashboards (Artifact, Signal, Context Handler) for **LLM workflow systems**, supporting daily operations for research and engineering teams.
- Implemented a **WebSocket-based real-time inference** latency optimization pattern, cutting update latency by **300ms** to improve visibility into live LLM execution states.
- Developed Apache **Airflow DAGs** with **Celery** distributed workers, handling 1,000+ batch jobs per worker with custom deduplication logic and tested using **Vite Jest** and other unit testing.
- Integrated OpenAI Api in dashboard workflows, prompt engineering to improve inference reliability, output quality for research teams.

Krishlabs | Junior Full-Stack Developer (Contract) | Bangalore, India |

Nov 2024 – Apr 2025

- Architected** a responsive **Next.js** and **Express.js** marketing platform, integrating **Framer Motion** and validated forms, increasing engagement by **25%** and sign-ups by **40%**.
- Engineered a cross-platform social media application using **Expo React Native**, **Node.js**, **Express.js**, and **MongoDB**, optimizing performance throughput and reducing latency by **60%** through **Socket.IO**.
- Deployed **15+ REST API** endpoints on **AWS EC2** with **S3** media storage and **MongoDB**, maintaining **99.8%** uptime for authentication and content services.

OPEN SOURCE CONTRIBUTIONS

- [louislam/uptime-kuma](#), [lingdojo/kana-dojo](#): Authored dynamic gRPC port binding ([PR: fix/grpc-ports](#)), dashboard filter query param synchronization ([PR: feat/sync-filters](#), +241 lines); contributed visual theme ([PR](#)), configuring JSON for minimal UI. collaborated through 16+ code review discussions to improve test isolation and UX.

PROJECTS

Gaminai — AI Research Assistant | React • Express.js • TypeScript • Postgres • Prisma ORM | [Live Demo](#)

- Built a Retrieval-Augmented Generation (RAG) pipeline with exactly-once semantics and idempotent memory handling, integrating OpenAI/Anthropic APIs for a full-stack AI research assistant with authenticated chat history and live answer streaming.
- Applied prompt engineering and real-time web source retrieval to ground model responses, maintaining persistent conversation memory across sessions.

ProScope-Focus | YouTube API • React • IndexedDB • Express.js • OAuth 2.0 • Passport.js • Vercel | [Live](#)

- Applied **architectural judgment** in choosing **IndexedDB** for offline-first state management in a **React 19** app, utilizing a **600ms** debounced sync buffer to keep state updates under **100ms**.
- Developed an **Express.js** middleware layer with in-memory **caching** to optimize **YouTube** metadata retrieval.

RunAway | JavaScript, Canvas 2D, Node.js | [Live](#) | [Code](#)

- Integrated the YouTube IFrame Player API behind a custom transport UI, plus a procedural Web Audio synthesizer.
- Achieved **30+ daily users** and **48+ unique visitors**, demonstrating consistent organic engagement.

ACHIEVEMENTS & CERTIFICATIONS

- Hackathon:** 3rd place, Unstop Powered Hackathon (200+ participants).
- Certifications:** JS & Java (IIT Bombay, 90% score). **Ongoing:** DSA practice in Java [GitHub](#).